

Christy Biji

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Education

Whiting School of Engineering, Johns Hopkins University

Aug 2021 - Jan 2023

MS in Chemical and Biomolecular Engineering

- **Thesis:** Development of a human iPSC derived microglia assay for evaluation of target dependent activation and pharmacological inhibition ([Link](#) 🔗)
- **Relevant Courses:**

Department of Biotechnology, Savitribhai Phule Pune University

Oct 2020

B.Tech Biotechnology (Sinhgad College of Engineering)

- **Thesis:** Extraction & Application of Hydrophobin ([Link](#) 🔗)

Research Experience

The Broad Institute of MIT & Harvard

Cambridge, MA

Research Associate II, Center for the Development of Therapeutics (CDoT)

April 2023 – Present

Current Projects

- Generating a large-scale morphological dataset via Cell Painting for a machine learning pipeline designed to identify drugs that revert pediatric cancer cells to a healthy phenotype.
Collaboration: Dr. Gregory Way, University of Colorado. 🔗
- Developing and validating genetically engineered HiBit cell lines to serve as the basis for a novel, high-throughput chromatin displacement assay for a PAX3-FOXO1 targeted drug discovery campaign in rhabdomyosarcoma. *Collaboration: Dr. Angela Koehler, MIT.*
- Enabling a hit-to-lead campaign and the discovery of novel agonist & positive allosteric modulators (PAMs) by screening 1200+ compounds against multiple GPCR targets (C3aR, GPR34).
Collaboration: Stanley Center, The Broad Institute

Completed Projects

- Executed a data-driven drug repurposing screen that validated CDK9 inhibition as a novel therapeutic strategy for TFE3 fusion-driven renal cell carcinoma *Collaboration: Dr. Viswanathan, DFCI.*
- Conducted a high-throughput differential viability screen against CALR-mutant neoplasms, delivering 65 validated, selective hits from a library of 40k compounds to launch a new therapeutic program. *Collaboration: Dr. Jonas Jutzi, BWH & Dr. Aaron Robida, University of Michigan.*
- Implemented GSK3 β biochemical assay to drive the structure-activity relationship (SAR) for a medicinal chemistry campaign targeting bipolar disorder. *Collaboration: Stanley Center, The Broad Institute*
- Led initial assay development for a multi-assay biophysical platform (FP, ITC, SPR) designed to quantify protein-peptide interactions for an HPV E6 drug discovery campaign.
Collaboration: P2P Initiative & Sellers Lab.
- Generated morphological profiles of novel BMP pathway modulators via Cell Painting to enable mechanism of action (MoA) studies *Collaboration: Dr. Dennis Schade, University of Kiel & Dr. Anne Carpenter, The Broad Institute.*
- Profiled morphological changes in cancer cells under hyperoxia-induced stress using Cell Painting.
Collaboration: Dr. Peter Tsvetkov, Harvard University.

Merck

Master Thesis Researcher, Discovery Neuroscience

Boston, MA

June 2022 – Jan 2023

- Led a project to establish iPSC-derived microglia platform to model CNS-specific NLRP3 inflammasome activation in neurodegenerative diseases.
- Characterized the dose-dependent NLRP3 inhibition with the small molecule MCC950, establishing IC50 values in CNS (iPSC-microglia) and peripheral (THP-1, PBMC) cell lines.
- Quantified inflammasome activation by measuring IL-1 β secretion via immunoassay and visualized ASC speck assembly using immunocytochemistry.

The Brady Urological Institute at Johns Hopkins

Graduate Researcher, Part-time

Baltimore, MD

Nov 2021 – May 2022

- Investigated host immune response to novel bioengineered bladder tissues by optimizing 3D gelatin scaffold fabrication via systematic physicochemical characterization
- Quantified cellular infiltration and spatial distribution in 3D constructs through quantitative analysis of histology and immunofluorescence imaging data.

Bayview Medical Centre at Johns Hopkins

Graduate Biospecimen Manager, Part-time

Baltimore, MD

Nov 2021 – May 2022

- Managed a clinical biorepository for neurological research, leveraging a **LIMS** (Freezerpro) to ensure the integrity of key samples (CSF, plasma, serum) and enhance data quality.

Publications

Cancer target discovery enabled by transcriptome-based virtual CRISPR screening

(on *bioRxiv*) [🔗](#) [Under Review @ Science]

Sadagopan A, Li B, Li J, Cui Y, Deodhar R, Yang D, Wu Y, Konda P, **Biji C**, Thapa DR, Thakur M, Weiss CN, Choueiri TK, Cheah JH, Doench JG, Drapkin BJ, Viswanathan SR. *bioRxiv* [Preprint]. 2025

GPR34 regulation of disease-associated microglial states and responses to physiological stimuli (on *bioRxiv*) [🔗](#)

Geller A, Kwon MJ, Simmons SK, Xu Q, Martenis WE, Doman J, Natarajan S, Stalnaker KJ, Zhang J, Adiconis X, Nguyen T, Demers M, Batzli D, Valle-Tojeiro A, **Biji C**, Aryal S, Pribiag H, Depp C, Morshed N, Misri D, Bohannon D, Longhi E, Keshishian H, Carr SA, McKinney D, Gould AE, Lebois E, Weiwer M, Johnson M, Levin JZ, Zhang YL, Sheng M, Kunwar PS. *bioRxiv* [Preprint]. 2025

Poster Presentation

Biji C, Tomkinson J, Curd J, Skepner A, Way G and Cheah J. *Generation of a Cell Morphological Landscape in Pediatric Cancer Cell Lines*. Broad Retreat, December 2025, Cambridge, MA.

Geller A, McKinney D., Longhi E, **Biji, C.**, et al. *Developing brain penetrant activators of microglia specific GPR34 for modulating function and treating neurodegenerative and psychiatric diseases* Stanley Center Symposium, September 2025, Cambridge, MA.

Weiwer M., Kyei-Baffour K., Wagner F, Shekar M, **Biji, C.**, et al. *Inhibition of GSK3 α,β rescues motoric and cognitive phenotypes in a preclinical mouse model of CTNNB1 Syndrome*. Broad Retreat, December 2024, Cambridge, MA.

Computational & Quantitative Skills

- **Machine Learning & Statistical Modeling:** Classification (Logistic Regression, Random Forests), clustering (k-means, Hierarchical), dimensionality reduction (PCA, UMAP), statistical inference, and hypothesis testing, Python (Pandas, Scikit-learn, Matplotlib), R (Tidyverse), MATLAB, Java, Git version control, and Terra cloud computing.
- **Bioimage & High-Content Data Analysis:** End-to-end experience in high-content screening workflows, including quantitative morphological profiling (CellProfiler, ImageJ, Cellpose) and large-scale data analysis (Genedata Screener, LIMS).
- **Experimental Biology Foundation:** Mammalian cell culture, Cell Painting, ELISA, High Throughput multiplexed screening in 384-well format, Dose-response profiling, immunocytochemistry (ICC), BSL2+/BSL2 and with automated screening instrumentation (Opera Phenix High Content Screening System, Beckman Coulter Echo, Biotek, Pherastar).

Projects

Knockout NEU3 gene target with CRISPR-Cas9

[Project Report](#) 

(Cell & Tissue Engineering Lab)

- Engineered a CRISPR-Cas9 gene-editing tool to knock out the sialidase Neu3 in CHO cells, a strategy aimed at increasing the sialylation and therapeutic half-life of recombinant hIFN γ .

Engineering Lungs with mtDNA-Edited iPSCs

[Technical Report](#) 

(Tissue Engineering Project)

- Proposed a novel framework for lung bioengineering combining decellularized scaffolds with autologous, **mtDNA-edited iPSCs** to prevent transplant rejection and expand the donor organ pool.

BoomExtractor - AI-Powered Scientific Workflow Visualization

[BoomExtractor](#) 

(Broad Hackathon 2025 ([Slides](#) ))

- **Awarded BroadHacks Best Overall Project and the People's Choice award.** ([Link](#) )
- Led the end-to-end development of an AI-driven tool to accelerate research comprehension by automatically generating scientifically-validated experimental workflow diagrams from papers using **Python**, **Gemini AI**, and **React**.

MIT-IDSS-Recommendation-System

[Project Link](#) 

(Course: Recommendation Systems)

- Engineered an end-to-end recommendation system by benchmarking collaborative filtering models (User-User, SVD) and optimizing the top-performing **SVD model** with **GridSearchCV** to achieve a final **RMSE of 0.88**.

MIT-IDSS-Lead-Scoring-Model

[Project Link](#) 

(Course: Classification and Hypothesis Testing)

- Developed an end-to-end lead scoring pipeline by tuning multiple classifiers (**Random Forest**, **Decision Tree**) with **GridSearchCV** to optimize for recall (82%), enabling data-driven prioritization of high-potential sales leads.

MIT IDSS Project: Customer Personality Segmentation

[Project Link](#) 

(Course: Making Sense of Unstructured Data)

- Applied unsupervised learning (**K-Means**, PCA) to segment a retail customer database, identifying and profiling distinct high-value and low-value personas to inform targeted marketing and

retention strategies.

Leadership and Services

Treasurer, Women at Broad <i>The Broad Institute</i>	Sep 2025
Spot Award - awarded for excellence in the service of the department <i>CDoT, The Broad Institute of MIT and Harvard</i>	May 2025
Graduate Student Liaison Committee <i>Whiting School of Engineering - Johns Hopkins University</i>	May 2021

Courses and Certification

Data Science and Machine Learning: Making Data-Driven Decisions <i>MIT Institute for Data, Systems, and Society (IDSS) (Link </i>	Jan 2025
Mathematical Statistics (Graduate-Level Coursework) <i>Harvard University Extension School Grade: A (Transcript Link </i>	Dec 2024
Introduction to the Biology of Cancer <i>Johns Hopkins University, Coursera (Link </i>	July 2020
Genes and the Human Condition (From Behavior to Biotechnology) <i>University of Maryland, Coursera (Link </i>	June 2020
Antimicrobial Resistance - Theory and Methods <i>Technical University of Denmark, Coursera (Link </i>	June 2020